AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

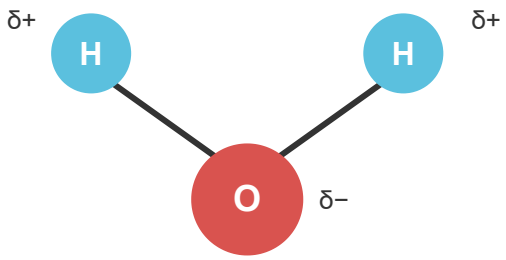
Chemistry for biologists

Living things are built from a small set of molecules, and most of their behaviour follows from the shapes of those molecules and the forces between them. This topic covers water, carbohydrates, lipids and proteins, and explains how the structure of each one fits its job in an organism.

1. Water: a polar molecule

A water molecule is made of one oxygen atom covalently bonded to two hydrogen atoms. The shared electrons are not held equally. Oxygen is much more electronegative than hydrogen, so it pulls the bonding electrons towards itself. The oxygen end becomes slightly negative (δ−) and the two hydrogen ends become slightly positive (δ+).

The molecule is bent rather than straight, at roughly 104.5 degrees. Because of this shape the small charges do not cancel out, so the whole molecule behaves as a dipole. This is the single most important fact about water, because nearly every property discussed below depends on it.



δ− on oxygen, δ+ on each hydrogen

Why water is polar

1. Oxygen has a higher electronegativity than hydrogen.
2. The shared bonding electrons spend more time near the oxygen nucleus.

   Electrons are pulled towards the atom that attracts them more strongly.
3. Oxygen becomes δ− and each hydrogen becomes δ+.
4. The molecule is bent, so the two bond dipoles add up instead of cancelling.

   A straight molecule with the same bonds would have no overall dipole.
5. The result is a permanent dipole, which lets water form hydrogen bonds with other polar molecules.

- **Covalent bond:** a shared pair of electrons between two atoms, the strong bond holding the O and H atoms of one water molecule together.
- **Hydrogen bond:** a weak attraction between a δ+ hydrogen on one molecule and a δ− oxygen or nitrogen on another. It is much weaker than a covalent bond but there are very many of them.
- **Polar molecule:** a molecule with an uneven charge distribution, so it has a positive end and a negative end.

2. Why water is essential for life

Water is the medium in which almost all biochemistry happens, and it is the most abundant compound in most cells. Its properties are not accidents. Hydrogen bonding between water molecules holds them together, and this gives water a high boiling point and a high specific heat capacity, and makes it act as a solvent, a transport medium and a thermal buffer.

Water as a solvent

Ionic compounds such as sodium chloride dissolve because water molecules surround the separated ions. The δ− oxygens cluster around Na+ and the δ+ hydrogens cluster around Cl−, each ion being wrapped in a shell of water and kept apart. Polar covalent molecules such as glucose dissolve in the same way, forming hydrogen bonds with water. Non-polar substances such as fats do not, because they cannot compete with the hydrogen bonds between water molecules. They are hydrophobic and separate out from the water.

This is why water is a good transport medium. Sugars, amino acids, ions and gases can be carried in the blood, the cytoplasm and the xylem and phloem sap because they are dissolved in water. Dissolved substances also stay available for reactions, because they can move and collide freely.

Cohesion, adhesion and surface tension

Water molecules are attracted to each other by hydrogen bonds, which is called cohesion. Molecules in the middle of a liquid are pulled equally in all directions. Molecules at the surface are pulled only sideways and downwards, so the surface is pulled tight. This is surface tension, which lets small insects such as pond skaters walk on the surface without breaking it.

Water is also attracted to other polar surfaces such as the walls of xylem vessels. This is adhesion. Cohesion keeps a continuous column of water together under tension during transpiration, and adhesion stops the column from pulling away from the vessel walls.

Specific heat capacity and latent heat

Specific heat capacity is the energy needed to raise the temperature of 1 kg of a substance by 1 °C. Water has a high value, about 4200 J kg^{-1} $°C^{-1}$. A large part of the energy input goes into breaking hydrogen bonds before the molecules can move faster, so the temperature rises slowly. This stabilises the temperature of lakes, seas and the cytoplasm of cells.

Latent heat of vaporisation is the energy needed to change 1 kg of a liquid into a gas at a constant temperature. Water has a high value because a lot of energy is needed to break the hydrogen bonds holding the liquid together.

When water evaporates from the skin, the most energetic molecules escape and the remaining water is cooler. This is evaporative cooling, the basis of sweating.

Property	Why it happens	Biological significance
High specific heat capacity	Energy is absorbed breaking hydrogen bonds	Stable temperature in organisms and aquatic habitats
High latent heat of vaporisation	Many hydrogen bonds must be broken to evaporate	Cooling by sweating and transpiration
High surface tension	Cohesion pulls surface molecules inwards	Insects walk on water; capillary action in plants
Density maximum at 4 °C; ice less dense than liquid	Hydrogen bonds hold molecules apart in ice	Ice floats, insulating the water below so aquatic life survives winter
Universal solvent	Polar and charged molecules are attracted to water	Transport and a medium for reactions

KEY IDEA

Ice is less dense than liquid water because hydrogen bonds hold the molecules in an open lattice. Most substances become denser when they freeze, but water does not, which is why ponds freeze from the top down.

COMMON MISTAKE

Do not confuse specific heat capacity with latent heat. Specific heat is about a temperature change within one state. Latent heat is about a change of state at constant temperature.

3. Carbohydrates

Carbohydrates contain carbon, hydrogen and oxygen, with hydrogen and oxygen in a 2:1 ratio. Their general formula is $C_n(H_2O)_n$. They are the main energy source for cell respiration and, as polymers, also serve as stores of energy. They are classified by the number of sugar units they contain: monosaccharides, disaccharides and polysaccharides.

Monosaccharides

Monosaccharides are the simplest sugars, with three to seven carbon atoms. The ones you need to know are glucose, fructose and galactose, all of which have the formula $C_6H_{12}O_6$ but different structures. Glucose exists in two ring forms, alpha and beta, which differ only in the position of the –OH group on carbon 1. Only alpha-glucose is needed for this topic, and it is the form that makes starch and glycogen. Monosaccharides are soluble, sweet and reducing sugars, and they are small enough to pass through cell membranes by facilitated diffusion or active transport.

EXAM TIP

When a question asks why glucose is soluble, give the reason in terms of the structure: the many –OH groups make glucose polar, so it forms hydrogen bonds with water.

Disaccharides and the glycosidic bond

Two monosaccharides join by a condensation reaction. An –OH group from one sugar combines with an –H from the other, and a molecule of water is released. The new covalent link is a glycosidic bond. Hydrolysis is the reverse: a water molecule is added to break the bond. Enzymes such as maltase, sucrase and lactase carry out these hydrolysis reactions during digestion.

- **Maltose:** two alpha-glucose units joined by a 1,4-glycosidic bond. It is formed when starch is broken down, for example during germination.
- **Sucrose:** alpha-glucose joined to fructose by a 1,2-glycosidic bond. It is the transport sugar in plants and the sugar most people eat.
- **Lactose:** galactose joined to glucose. It is the main sugar in milk and is hydrolysed by lactase.

Build maltose, and then a longer chain, one condensation reaction at a time.

Polysaccharides

Polysaccharides are long chains of many monosaccharide units joined by glycosidic bonds, typically several hundred to thousands of units long. Their size makes them insoluble, so they do not affect the water potential of the cell and cannot diffuse out of it. This is why they are good storage molecules: a cell can store a large amount of glucose as starch or glycogen without a huge osmotic effect.

Starch and glycogen

Starch is the storage polysaccharide of plants. It is a mixture of two polymers of alpha-glucose. Amylose is an unbranched chain of 1,4-linked units that coils into a helix, which makes it compact. Amylopectin is branched, with 1,6-glycosidic bonds appearing about every 20 to 30 glucose units, and the branches end in many terminal glucose units.

Glycogen is the storage polysaccharide of animals and fungi. Its structure is like amylopectin but with more frequent branches, roughly every 8 to 10 units. Many branches mean many free ends, so enzymes can add or remove glucose from many points at once. Glycogen can therefore be broken down quickly, which suits animal muscle and liver during intense activity.

Why glycogen suits animal energy storage

1. Glycogen is a polymer of alpha-glucose, so it stores a lot of energy in a small volume.

2. It is insoluble and does not create an osmotic effect, so the cell does not take in extra water.

   Dissolved glucose molecules would each add to the solute concentration.

3. Its many branches give many non-reducing ends.

4. Enzymes can release glucose from every end at once, so glucose is released rapidly when needed.

   Rapid glucose release supports the high demand for respiration during exercise.

Feature	Monosaccharide	Disaccharide	Polysaccharide
Number of units	1	2	Many (often hundreds or more)
Glycosidic bonds	None	1	Many
Solubility	Soluble	Soluble	Insoluble
Sweet taste	Yes	Yes	No
Osmotic effect	Yes	Yes	Little or none
Examples	Glucose, fructose, galactose	Maltose, sucrose, lactose	Starch, glycogen

4. Lipids

Lipids are a varied group of molecules that are insoluble in water and soluble in organic solvents. They are not polymers, because they are not made of many identical repeating units. They contain much less oxygen than carbohydrates, so when they are oxidised they release much more energy per gram. The main groups are triglycerides, phospholipids and steroids such as cholesterol.

Triglycerides

A triglyceride is made from one glycerol molecule and three fatty acids. Each fatty acid has a carboxyl group at one end and a long hydrocarbon tail. The carboxyl group of each fatty acid reacts with an –OH group of glycerol in a condensation reaction, forming an ester bond. Three water molecules are released. Hydrolysis of an ester bond reverses this, which is how lipase digests fat.

Saturated and unsaturated fatty acids

In a saturated fatty acid all the carbon–carbon bonds in the tail are single bonds, so the tail is straight. Straight tails pack closely together, with many van der Waals forces between them, so fats made from them are solid at room temperature. This is why butter is solid.

An unsaturated fatty acid has at least one carbon–carbon double bond. A double bond introduces a kink, so the tails cannot pack tightly. Fewer intermolecular forces act between them, so less energy is needed to separate them, and oils are liquid at room temperature. A polyunsaturated fat has more than one double bond.

COMMON MISTAKE

A double bond between carbons does not mean extra hydrogens are present. Unsaturated means there is less hydrogen than the maximum possible. The important point for melting point is the kink, not the hydrogen count.

Phospholipids and functions of lipids

A phospholipid has two fatty acid tails and a phosphate group in place of the third fatty acid. The phosphate head is polar and hydrophilic, while the tails are non-polar and hydrophobic. This dual character is what makes phospholipids form the bilayer of cell membranes, which is covered in more detail in the cell membrane topic.

The main functions of lipids are:

- **Energy storage:** a gram of fat releases about 9 kcal when oxidised, compared with about 4 kcal from a gram of carbohydrate or protein, because fat has far more C–H bonds and less oxygen.
- **Insulation:** fat under the skin reduces heat loss, and whale blubber is a good example.
- **Protection:** fat around vital organs cushions them from physical damage.
- **Buoyancy:** lipids are less dense than water, which helps aquatic animals float.
- **Waterproofing:** waxes and cuticles on leaves and skin stop water loss.

Why saturated fats are solid at room temperature

1. A saturated fatty acid tail has only single bonds, so it is straight.
2. Straight tails lie close together and their van der Waals forces add up across the whole length.
3. A lot of energy is needed to pull the tails apart.

   More energy is needed to overcome the forces, so the melting point is higher.
4. At room temperature the fat stays solid.

5. Proteins

Proteins are polymers of amino acids and are found in every cell. They do many jobs: enzymes catalyse reactions, transport proteins move substances, structural proteins give support, and antibodies defend the body. What a protein does depends on its three-dimensional shape, and that shape comes from the order of its amino acids.

Amino acids and the peptide bond

Every amino acid has the same core: a central carbon attached to an amine group (–NH_2), a carboxyl group (–COOH), a hydrogen and a variable R group. The R group differs between the twenty common amino acids and determines the properties of each one.

Amino acids join by condensation. The –COOH group of one amino acid reacts with the –NH_2 group of the next, forming a peptide bond (–CO–NH–) and releasing water. A chain of many amino acids joined this way is a polypeptide. Hydrolysis of peptide bonds breaks the chain again.

Four levels of protein structure

- **Primary structure:** the sequence of amino acids in the chain. This is determined by the gene coding for the protein.
- **Secondary structure:** local folding of the chain into an alpha helix or a beta pleated sheet, held in place by hydrogen bonds between the C=O of one residue and the N–H of another. These bonds are along the backbone, not the R groups.
- **Tertiary structure:** the overall three-dimensional folding, stabilised by interactions between R groups. These include ionic bonds, hydrogen bonds, disulphide bridges between cysteine residues and hydrophobic interactions.
- **Quaternary structure:** two or more polypeptide chains held together, as in haemoglobin.

Why changing one amino acid can change a protein's function

1. Changing one amino acid changes the R group at that position.
2. The new R group may have different charge, size or polarity.

   Only the R group varies, so the change is in its chemistry.
3. This alters the bonds that form between R groups, so the tertiary structure changes.
4. The shape of the active or binding site changes, so the protein may no longer bind its substrate.

Fibrous and globular proteins

Globular proteins are folded into compact, roughly spherical shapes. Their hydrophilic R groups point outwards, so they dissolve in water, and they can carry substances or act as enzymes. Haemoglobin is a globular protein with four subunits, each containing a haem group. Fibrous proteins have long chains arranged in parallel and held together by many cross-links. They are insoluble and strong, which suits structural roles. Collagen is a fibrous protein with three polypeptide chains wound into a triple helix. Each strand is coiled tightly, and this gives collagen a high tensile strength for tendons and skin.

Feature	Globular protein	Fibrous protein
Overall shape	Compact, roughly spherical	Long and thread-like
Solubility in water	Soluble	Insoluble
Main role	Transport, enzymes, signalling	Structural support
Example	Haemoglobin, insulin	Collagen, keratin

Denaturation

Heat, extreme pH, high salt concentration and some organic solvents break the hydrogen bonds, ionic bonds and disulphide bridges that maintain the tertiary and secondary structure. The chain unfolds and the shape of the active site is lost. The primary structure is not broken, because peptide bonds are unaffected, so the protein is denatured but the sequence stays the same. Denaturation is usually irreversible for enzymes.

KEY IDEA

Protein function depends on shape, shape depends on the sequence and on the bonds between R groups, and denaturation destroys the shape without breaking the peptide bonds.

EXAM TIP

When you explain a protein's shape, name the bond type and the level of structure. Saying "the bonds" is not enough for full marks.

Quick checks

QUICK CHECK

Which property of water is mainly responsible for the cooling effect of sweating?

- High specific heat capacity
- High latent heat of vaporisation
- High density at 4 °C
- Universal solvent

Sweat evaporates, and evaporation needs a large amount of energy to break hydrogen bonds. The energy comes from the skin, so the skin cools. Specific heat capacity describes temperature change inside a liquid, so it is the commonest wrong answer.

QUICK CHECK

A disaccharide is formed from two alpha-glucose molecules. What is released when the glycosidic bond forms?

- Carbon dioxide
- Oxygen
- Water
- A hydrogen molecule

Condensation joins two monosaccharides by combining an –OH group with an –H, and the two form water. Hydrolysis is the reverse. Students often think a bond forms by losing hydrogen gas, which is incorrect.

QUICK CHECK

Why is glycogen a better energy store in animal muscle than a single glucose molecule?

- It has a higher solubility in water
- It releases glucose more slowly
- It is insoluble and has many ends for rapid glucose release
- It contains more oxygen than glucose

Glycogen is insoluble, so it does not create a large osmotic effect, and its branches give many terminal glucose units for enzymes to act on at once. A common wrong answer is that glycogen is more soluble; it is actually the opposite.

QUICK CHECK

Which change would most likely lower the melting point of a fat?

- Making the fatty acid tails longer and fully saturated
- Introducing carbon–carbon double bonds into the fatty acid tails
- Joining the fatty acids to a larger glycerol molecule
- Increasing the number of hydrogen atoms in the tails

Double bonds create kinks that stop the tails packing together, which weakens the van der Waals forces and lowers the melting point. Longer saturated tails would pack more tightly and raise the melting point. Adding hydrogen to a tail is saturation, which raises the melting point.

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Mammalian transport

Mammals are large and very active, so diffusion alone cannot supply their cells. This topic explains the double circulatory system, the structure and cycle of the heart, the three types of blood vessel, the formation of tissue fluid, and how blood carries oxygen and carbon dioxide.

1. Why organisms need a transport system

Every cell needs oxygen and nutrients and must remove carbon dioxide and waste. Small, flat or thin organisms can exchange these by diffusion across their surface. The rate of diffusion depends on the surface area, the concentration gradient and the distance. As an organism gets larger, its surface area to volume ratio falls, while its metabolic demand rises with its volume. Diffusion distances also grow, so diffusion becomes far too slow to meet demand.

A transport system solves this by moving materials in bulk, by mass flow, over long distances. Blood is pumped under pressure through a network of vessels, so most substances travel quickly to the cells and only the final short distance is covered by diffusion.

KEY IDEA

Mass flow is the bulk movement of a fluid carrying dissolved substances. Diffusion is the slow movement of individual molecules down a gradient. A transport system relies on mass flow over long distances and diffusion only over short ones.

2. Types of circulatory system

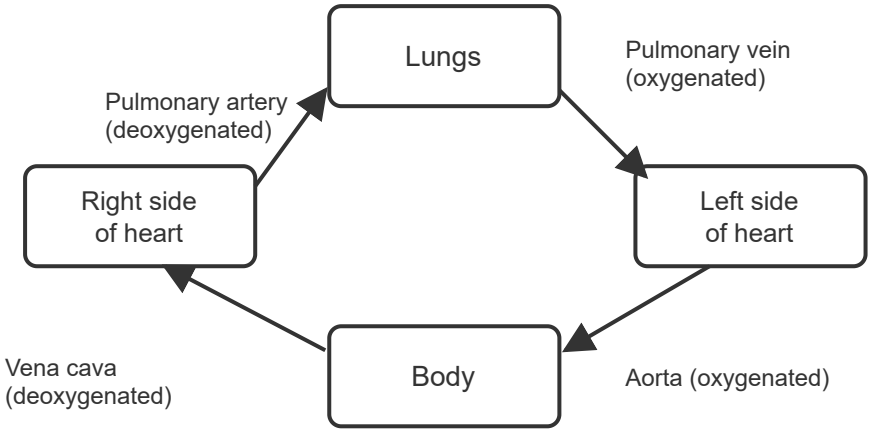
In an open circulatory system, such as that of insects, blood is pumped into body cavities and bathes the organs directly. Pressure is low and flow is slow, which is acceptable because these animals have a low metabolic rate and small body size. In a closed system, blood always stays inside vessels. The vessels carry it at higher pressure, so mass flow is faster. All vertebrates have a closed system.

Fish have a single circulation. Blood goes from the heart to the gills, from the gills to the body, and back to the heart. Blood loses much of its pressure passing through the capillaries of the gills, so the flow to the body is slow. Mammals have a double circulation. Blood passes through the heart twice in each full circuit, once on the pulmonary route and once on the systemic route.

Why double circulation suits mammals

- **Separate pumps:** the right side of the heart sends blood to the lungs, and the left side sends blood to the body. Each side can work at a pressure suited to its circuit.

- **Pressure control:** the lungs have delicate, thin-walled capillaries, so blood is pumped to them at low pressure. The body needs high pressure to deliver oxygen quickly.
- **No mixing:** oxygenated and deoxygenated blood stay separate, which keeps a steep concentration gradient for oxygen between blood and tissues.



3. Structure of the mammalian heart

The heart is a muscular pump about the size of a fist. It has four chambers. The two atria are thin-walled and receive blood from the veins. The two ventricles are thick-walled and push blood out into the arteries. The left and right sides are separated by a septum, so oxygenated and deoxygenated blood never meet. The heart is enclosed in the pericardium, a membrane with a small amount of fluid that reduces friction as the heart beats.

The wall has three layers. The epicardium is the outer protective layer. The myocardium is the thick cardiac muscle that contracts. The endocardium is the smooth inner lining that prevents clotting and reduces friction on the blood.

The left ventricle wall is about three times thicker than the right. The left ventricle must generate enough pressure to push blood through the whole systemic circuit, which has high resistance. The right ventricle only has to push blood through the nearby lungs, where high pressure would damage the thin capillaries.

Heart valves

Valves ensure blood flows in one direction. There are two kinds. The atrioventricular valves lie between each atrium and ventricle. The bicuspid (mitral) valve on the left has two flaps, and the tricuspid valve on the right has three. The semilunar valves lie at the base of the aorta and the pulmonary artery, and each has three cup-shaped pockets. Semilunar valves stop blood flowing back from the arteries into the ventricles.

4. The cardiac cycle

The cardiac cycle is one complete sequence of contraction and relaxation of the heart. It has two main phases. In systole the muscle contracts and pressure rises. In diastole the muscle relaxes and the chambers fill. The cycle is initiated by the sinoatrial node (SAN), a group of specialised cardiac muscle cells in the right atrium. The SAN generates an electrical impulse that spreads through both atria, making them contract together. The impulse reaches the atrioventricular node (AVN), which delays it briefly so the atria finish emptying before the ventricles

contract. The impulse then passes down the bundle of His and spreads through the Purkinje fibres, so the ventricles contract from the apex upwards.

One cardiac cycle, in order

1. Atrial systole: the atria contract and push blood through the open atrioventricular valves into the ventricles.

   Extra filling gives the ventricles a little more blood before they contract.

2. Ventricular systole begins: pressure in the ventricles rises above the atrial pressure.

   The rise in pressure shuts the atrioventricular valves, which stops backflow into the atria.

3. Pressure rises further, until it exceeds the pressure in the aorta and pulmonary artery.

   The semilunar valves are forced open, so blood leaves the heart.

4. Ventricular diastole: the ventricles relax, and pressure in them falls below the pressure in the arteries.

   The semilunar valves snap shut, so blood cannot flow back into the ventricles.

5. When ventricular pressure falls below atrial pressure, the atrioventricular valves open and the ventricles fill passively from the atria.

   The filling is driven by the pressure difference, not by muscle contraction.

COMMON MISTAKE

Valves open and close because of pressure differences, not because they are pulled by muscle. A valve opens when the pressure behind it is higher than the pressure in front of it.

Faulty valves and holes in the septum

If an atrioventricular valve does not close properly, blood leaks back into the atrium during ventricular systole. This is regurgitation. The heart has to work harder, and less blood is pumped forward, so the body receives less oxygen. Back-pressure can also raise pressure in the lungs. A hole in the septum mixes oxygenated and deoxygenated blood, so less oxygen reaches the tissues, and the heart has to pump extra blood to make up the shortfall.

5. Blood vessels

The structure of each vessel matches the pressure it carries. Arteries carry blood away from the heart at high pressure. Their thick walls contain elastic fibres that stretch during systole and recoil during diastole, which smooths the flow. Veins carry blood back to the heart at low pressure. Their walls are thinner, with less elastic tissue, and their lumens are wide. Veins contain valves that stop blood flowing backwards, and skeletal muscle movement squeezes blood along them. Capillaries are very narrow vessels whose walls are only one cell thick. This short diffusion distance allows exchange with the tissues, and their large total cross-sectional area slows the blood.

Feature	Artery	Capillary	Vein
Direction of flow	Away from heart	Between arteries and veins	Towards heart
Pressure	High, pulsing	Low	Low
Wall	Thick, muscular and elastic	One cell thick	Thin, less elastic
Lumen	Narrow	Very narrow	Wide
Valves	None (except heart)	None	Present

6. Tissue fluid

Tissue fluid bathes the cells and is the medium through which they exchange materials with the blood. It forms by filtration at the arterial end of a capillary. Hydrostatic pressure, the pressure of the blood itself, is high at this end because the blood has just left the heart. This pressure pushes water and small solutes such as glucose, oxygen and salts out through the capillary walls. Large plasma proteins are too big to leave and stay in the blood.

The plasma proteins lower the water potential of the blood, which creates an oncotic pressure that draws water back in. At the venous end the hydrostatic pressure has fallen, because much of it was lost on the way along the capillary. The oncotic pressure now exceeds the hydrostatic pressure, so water moves back into the capillary by osmosis. Most of the tissue fluid returns to the blood this way, and the rest drains into the lymphatic system, which returns it to the circulation.

Net movement of fluid across a capillary

1. At the arterial end, hydrostatic pressure is greater than the oncotic pressure.

   The blood has just been pumped out of the heart, so it is at high pressure.

2. The net pressure is outwards, so fluid is forced into the tissues.

3. Along the capillary, hydrostatic pressure falls.

   Friction and fluid loss lower the pressure as the blood moves along.

4. At the venous end, oncotic pressure is greater than hydrostatic pressure, so the net pressure is inwards.

5. Fluid returns to the capillary, and the excess drains into lymph vessels.

The diagram below compares the two pressures all the way along a capillary.

EXAM TIP

Explain tissue fluid formation by comparing two pressures at each end of the capillary. Stating only that pressure forces fluid out loses marks.

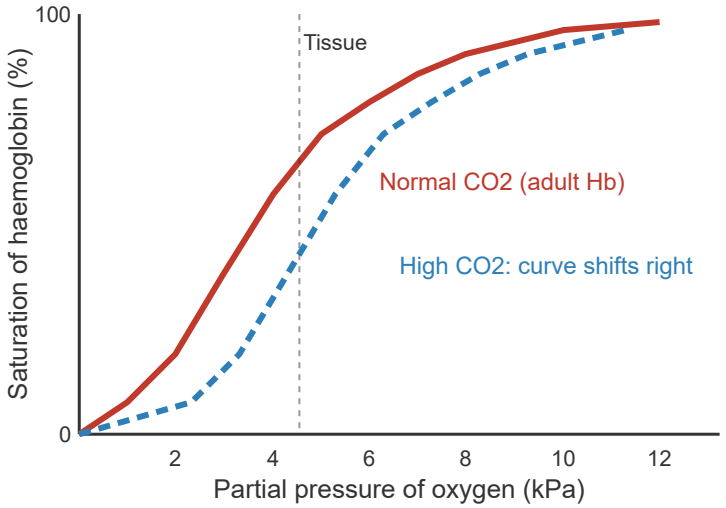
7. Blood clotting

When a vessel is damaged, platelets stick to the exposed collagen and to each other, forming a plug. Clotting factors then produce a cascade of reactions. Damaged tissue releases an enzyme that converts prothrombin into thrombin. This step needs calcium ions, and vitamin K is needed by the liver to make prothrombin. Thrombin converts soluble fibrinogen into insoluble fibrin, which forms a mesh that traps red blood cells and seals the wound. The cascade is tightly controlled, so clots form only where they are needed.

8. Transport of oxygen by haemoglobin

Haemoglobin is a globular protein in red blood cells, with four polypeptide subunits. Each subunit contains a haem group, which holds one iron ion (Fe^{2+}). Each iron ion can bind one oxygen molecule, so one haemoglobin molecule can carry four oxygen molecules. Binding is reversible, which is what allows loading in the lungs and unloading in the tissues.

Binding is cooperative. When the first oxygen binds, the shape of the haemoglobin changes so that the next oxygen binds more easily. The last oxygen binds several hundred times faster than the first, but once it is bound, oxygen is hard to remove. Because of this, the graph of percentage saturation against oxygen partial pressure is sigmoidal (S-shaped), called the oxygen dissociation curve.



Adult haemoglobin binds oxygen strongly at high partial pressure in the lungs, and the curve shifts right when carbon dioxide is high in the tissues.

Reading the curve

In the lungs, oxygen partial pressure is high, so haemoglobin is nearly fully saturated, and the flat top of the curve means a small drop in pressure changes little. In the respiring tissues, partial pressure is low. Here the steep middle section of the curve means a small fall in pressure releases a large amount of oxygen. The curve therefore helps haemoglobin load fully in the lungs and unload effectively in the tissues.

The Bohr effect

During activity, respiring tissues produce carbon dioxide. Carbon dioxide dissolves in the blood and forms carbonic acid, which lowers the pH. Lower pH changes the shape of haemoglobin so that it binds oxygen less tightly. The curve moves to the right, meaning that at any given oxygen partial pressure, less oxygen is bound. More oxygen is released where it is needed. This shift is called the Bohr effect.

KEY IDEA

A rightward shift of the oxygen dissociation curve means lower affinity for oxygen. Higher CO2 or lower pH causes this shift and so increases oxygen unloading in active tissues.

Fetal haemoglobin

Fetal haemoglobin has a higher affinity for oxygen than adult haemoglobin, so its curve lies to the left. The placenta is where oxygen passes from the mother's blood to the fetal blood. Because the fetal blood has a higher affinity, it can take oxygen from the maternal blood even when the oxygen partial pressure there is low. This is an important adaptation for development before birth.

Use the curve below to read off saturations for yourself.

9. Transport of carbon dioxide

Carbon dioxide is produced by respiration in the tissues and must be removed. It is carried in the blood in three forms. A small fraction, about 5%, dissolves directly in the plasma. A further 10 to 20% binds to the amino groups of haemoglobin to form carbaminohaemoglobin. Most of the carbon dioxide, roughly three quarters, is converted into hydrogencarbonate ions (HCO_3^-) inside the red blood cells.

Converting carbon dioxide to hydrogencarbonate in a red blood cell

1. Carbon dioxide diffuses from the tissue fluid into the red blood cell.

   The concentration of carbon dioxide is higher outside the cell, so it moves in down the gradient.

2. Carbon dioxide combines with water to form carbonic acid, H_2CO_3.

   The reaction is catalysed by the enzyme carbonic anhydrase inside the red blood cell, making it very fast.

3. Carbonic acid dissociates into hydrogen ions and hydrogencarbonate ions.

4. Hydrogencarbonate ions diffuse out into the plasma, and chloride ions move into the cell to balance the charge.

   This exchange is called the chloride shift, and it keeps the electrical charge balanced.

5. The hydrogen ions bind to haemoglobin, which helps release oxygen (the Bohr effect).

In the lungs, the process runs in reverse. Hydrogencarbonate ions re-enter the red blood cell and combine with hydrogen ions released from reduced haemoglobin to form carbonic acid. Carbonic anhydrase converts this to carbon dioxide and water, and the carbon dioxide diffuses into the alveoli to be breathed out.

Form of carbon dioxide	Approximate share	Where it happens
Dissolved in plasma	About 5%	Plasma
Carbaminohaemoglobin	10 to 20%	Bound to haemoglobin
Hydrogencarbonate ions	Most of the rest (about 75 to 80%)	Formed inside red blood cells, then in plasma

COMMON MISTAKE

Carbonic anhydrase speeds up the reaction between carbon dioxide and water. It does not change the position of equilibrium, and it does not produce any extra carbon dioxide.

Quick checks

QUICK CHECK

Why is the wall of the left ventricle thicker than the wall of the right ventricle?

- It has to hold more blood than the right ventricle
- It contains more valves
- It must generate higher pressure to pump blood round the whole systemic circuit
- It needs more oxygen than the lungs

The left ventricle pumps blood through the high-resistance systemic circuit, so it needs to generate high pressure. The right ventricle pumps only to the nearby lungs, where lower pressure protects the capillaries. The commonest wrong answer is about volume; both ventricles pump roughly the same volume per beat.

QUICK CHECK

During ventricular systole, which valves are open?

- Both atrioventricular valves
- Neither set of valves
- The atrioventricular valves only
- The semilunar valves

In systole, pressure in the ventricles rises above the pressure in the arteries, which forces the semilunar valves open. The atrioventricular valves are shut to stop backflow into the atria. Students often assume the valves that open during filling are open at the same time as the ones pushing blood out.

QUICK CHECK

In active muscle, the oxygen dissociation curve shifts to the right. What does this mean for oxygen delivery?

- Haemoglobin binds oxygen more tightly
- Haemoglobin binds oxygen less tightly, so more oxygen is released to the tissue
- Haemoglobin becomes more saturated at all partial pressures
- The lungs load less oxygen

A right shift means lower affinity: at any oxygen partial pressure less oxygen is bound, so more is released where it is needed. The commonest error is to read a right shift as higher affinity, which is the opposite of what happens.

QUICK CHECK

What is the main role of the enzyme carbonic anhydrase in red blood cells?

- It binds oxygen to haemoglobin
- It breaks down haemoglobin
- It transports chloride ions out of the cell
- It speeds up the reaction between carbon dioxide and water

Carbonic anhydrase catalyses the formation of carbonic acid from carbon dioxide and water in the tissues, and the reverse reaction in the lungs. Chloride shift is a passive exchange, not carried out by this enzyme.

AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

Cardiovascular health

Cardiovascular disease (CVD) covers conditions of the heart and blood vessels, most of which begin with the slow build-up of fatty plaques in the arteries. This topic explains how atherosclerosis develops, which lifestyle and genetic factors raise the risk, how obesity and cholesterol are measured, and how medicines and surgery treat the disease.

1. Cardiovascular disease and what it damages

Cardiovascular disease is any disease of the heart or blood vessels that reduces the supply of blood, and so oxygen and nutrients, to the tissues. Most CVD is caused by atherosclerosis, in which plaques build up inside the walls of arteries. The plaque narrows the lumen, which raises blood pressure and reduces flow. If the artery becomes completely blocked, the tissue it supplies loses its oxygen supply and dies.

The main forms of CVD are:

- **Coronary heart disease (CHD):** narrowing of the coronary arteries that supply the heart muscle itself.
- **Angina:** chest pain caused by a temporary shortage of oxygen to the heart muscle, usually during exertion, which eases with rest.
- **Myocardial infarction (heart attack):** a coronary artery is blocked, and the heart muscle it supplies dies from lack of oxygen.
- **Stroke:** loss of brain function caused by a blocked artery (ischaemic stroke) or a burst artery (haemorrhagic stroke) supplying the brain.
- **Aneurysm:** a weakened, bulging section of an artery wall, which may rupture and cause severe internal bleeding.

2. How atherosclerosis develops

Atherosclerosis begins with damage to the endothelium, the single layer of cells lining the artery. The endothelium is damaged by high blood pressure, by chemicals in cigarette smoke, by high concentrations of LDL cholesterol in the blood, and by oxidative damage from free radicals. Damage triggers an inflammatory response. White blood cells, particularly macrophages, move into the wall of the artery and take up fats, including cholesterol from LDL. These cells become foam cells, which accumulate to form a fatty streak and then a plaque.

Over time the plaque enlarges, its cap can crack, and the artery wall loses its elasticity. The lumen narrows and blood pressure rises. A damaged plaque surface also activates platelets, which can form a clot (thrombus). A clot that grows large enough can block the artery completely.

From endothelial damage to a blocked artery

1. The endothelium is damaged, for example by high blood pressure or by smoke chemicals.
2. The damaged lining lets LDL cholesterol enter the artery wall.

   Damaged endothelium is more permeable, so LDL particles pass into the wall more easily.
3. Inflammation attracts white blood cells, which take up the cholesterol and become foam cells.
4. Foam cells and fibrous material build a plaque, which narrows the lumen.

   Narrowing raises the resistance to blood flow, so blood pressure rises further.
5. The plaque may rupture and trigger a clot, which can completely block the artery.

   Blood flow past the blockage stops, so the tissue beyond it is deprived of oxygen.

KEY IDEA

Atherosclerosis is a process that takes years. Damage to the endothelium is the starting point, and cholesterol deposition, inflammation and clot formation follow from it.

3. Risk factors

A risk factor is something that makes CVD more likely. Some risk factors cannot be changed, while others can be modified through lifestyle and treatment. Risk factors are usually linked: obesity, inactivity and high-fat diets often occur together and push blood pressure and cholesterol in the same direction.

Non-modifiable	Modifiable
Age	Diet
Gender	Smoking
Family history and genes	High blood pressure
	High blood cholesterol (especially LDL)
	Obesity and high body mass index
	Physical inactivity
	Stress
	Excess alcohol

Age and gender

The risk of CVD increases with age in both men and women. With age, arteries lose elasticity and become stiffer, so the pressure needed to push blood through them rises. Men have a higher risk than women at younger ages.

Oestrogen, present in women before the menopause, helps keep LDL low and HDL high, and it reduces plaque build-up. After the menopause, the protective effect falls and the difference between the sexes narrows.

Diet and cholesterol

Dietary fat affects blood cholesterol. Saturated fats, found in fatty meat and full-fat dairy, raise LDL. Unsaturated fats, such as vegetable oil, tend to lower LDL and contribute to HDL. Fish oil contains omega-3 fatty acids, which are associated with a lower risk of heart disease. Salt raises blood pressure because sodium retains water in the blood, increasing blood volume and the pressure it exerts on the artery walls. Fruit and vegetables contain antioxidants, such as vitamin C and polyphenols, which neutralise free radicals and protect the endothelium from oxidative damage.

EXAM TIP

When you discuss diet, say which type of fat, which lipoprotein it affects and how that affects the artery. A general statement that "fat is bad" will not earn full marks.

LDL and HDL

Cholesterol is carried in the blood by lipoproteins, which are particles of protein and lipid. Low-density lipoprotein (LDL) carries cholesterol from the liver to the tissues, and excess LDL deposits cholesterol in artery walls. High-density lipoprotein (HDL) carries cholesterol from the tissues back to the liver for breakdown. High LDL and low HDL together increase the risk of atherosclerosis.

Feature	LDL	HDL
Main role	Delivers cholesterol to tissues	Removes cholesterol from tissues to the liver
Effect on arteries	Promotes plaque formation	Helps reduce plaque
Linked fat type	Saturated fats	Unsaturated fats
Influenced by	Saturated fat intake, high blood cholesterol	Exercise and moderate alcohol raise it; smoking lowers it

Smoking

Cigarette smoke affects the cardiovascular system in several ways. Carbon monoxide binds to haemoglobin to form carboxyhaemoglobin, which is more stable than oxyhaemoglobin. This reduces the oxygen carried by the blood, so the heart and brain receive less oxygen and must increase heart rate to compensate. Nicotine stimulates the release of adrenaline, which raises heart rate and constricts arteries, raising blood pressure. Smoking also lowers HDL, promotes platelet activation and clot formation, and increases free radicals that damage the endothelium. Together these effects make atherosclerosis and thrombosis more likely.

How smoking raises the risk of CVD

1. Carbon monoxide forms carboxyhaemoglobin, which carries less oxygen.

   Less oxygen reaches the heart and brain, so they work harder to compensate.

2. Nicotine releases adrenaline, which raises heart rate and constricts arteries.

   Constricted arteries have higher resistance, so blood pressure rises.

3. Smoking lowers HDL and increases free radicals.

   Less HDL means less cholesterol is removed from the artery walls, and free radicals damage the endothelium.

4. Platelets become more active, so clots form more readily.

   A clot on a damaged plaque can block the artery and cause a heart attack or stroke.

Exercise, stress and alcohol

Physical activity lowers CVD risk. Exercise strengthens the heart muscle so that it contracts more forcefully and pumps more blood per beat, which lowers resting heart rate. It also dilates blood vessels, which lowers blood pressure, raises HDL and helps control weight. Exercise uses energy from fat as well as carbohydrate, so it helps balance energy intake and output.

Stress raises blood pressure and heart rate in the short term, and in some people it leads to overeating and obesity, which adds to the risk. Moderate alcohol intake is linked to higher HDL, and red wine contains antioxidants. However, heavy drinking raises blood pressure, contributes to obesity and irregular heartbeats, and damages the liver. Safe limits are around 3 to 4 units a day for men and 2 to 3 units for women.

4. Measuring obesity

Obesity is excess body fat, and it is a major risk factor for CVD because excess fat raises blood pressure and cholesterol. Two simple measurements are used to assess it.

Body mass index (BMI)

BMI is a quick way to check whether a person's weight is appropriate for their height. It is calculated by dividing body mass in kilograms by height in metres squared. Obesity-related health problems rise when BMI exceeds about 25, and a BMI of 30 or more is classed as obese.

Worked example: BMI for 95 kg and 1.70 m

1. Write the formula: BMI = mass ÷ height², with mass in kg and height in m.

2. Square the height: 1.70 × 1.70 = 2.89 m².

   Height is squared because body volume scales with the square of linear dimensions, so this corrects for taller people being heavier.

3. Divide the mass by the squared height: 95 ÷ 2.89 = 32.87.

4. Compare with the table: 32.87 lies between 30 and 40, so the person is classed as obese.

BMI (kg m^{-2})	Classification
Below 20	Underweight
20 to 24.9	Normal
25 to 29.9	Overweight
30 to 40	Obese
Above 40	Severely obese

COMMON MISTAKE

BMI does not distinguish fat from muscle. Athletes can have a high BMI with little body fat, and older people can have a normal BMI while losing muscle and gaining fat. It is also not reliable for children and pregnant women.

Body mass changes when the energy taken in as food and the energy used by the body are out of balance.

Waist-to-hip ratio

Fat stored around the abdomen, sometimes called central or android fat, is more closely linked to heart disease than fat on the hips and thighs. The waist-to-hip ratio measures this distribution. It is calculated by dividing the waist measurement by the hip measurement. A ratio above 0.80 for women or 0.95 for men indicates a raised risk of several diseases.

5. Treatment of cardiovascular disease

Treatment is chosen according to the severity of the disease and the risk factors present. It usually begins with lifestyle changes, and medicines or procedures are added when these are not enough. The aims are to lower blood pressure, reduce cholesterol, prevent clots and restore blood flow.

Lifestyle changes

Giving up smoking, eating less saturated fat and salt, eating more fruit and vegetables, taking regular exercise, keeping body weight in a healthy range and drinking alcohol within safe limits all reduce risk. These changes are the foundation of treatment because they target the causes rather than the symptoms.

Medicines that lower blood pressure

- **ACE inhibitors:** block angiotensin-converting enzyme, so less angiotensin II is made. Less angiotensin II means less vasoconstriction, so blood pressure falls. Side effects include a dry cough and dizziness.
- **Angiotensin receptor blockers:** block the receptors that angiotensin II acts on, so the arteries do not constrict in response. Side effects include dizziness and rash.
- **Beta blockers:** block adrenaline from binding to beta receptors in the heart. This slows the heart rate and reduces the force of contraction, so the output per minute falls and blood pressure drops.
- **Diuretics:** act on the kidneys to increase the volume of urine, removing excess salt and water. Blood volume falls, so less blood is pumped and pressure drops.
- **Calcium channel blockers:** stop calcium entering the cells of the heart and the walls of arteries. The arteries relax, and the heart works less hard.

Medicines that lower cholesterol or prevent clots

Statins are drugs that inhibit HMG-CoA reductase, the enzyme in the liver that makes cholesterol. Less cholesterol is made, so the liver takes more LDL from the blood, and LDL levels fall. Anticoagulants such as warfarin and heparin interfere with the clotting cascade, reducing the production of clotting factors. Antiplatelet drugs such as aspirin make platelets less sticky, so they are less likely to aggregate and form clots. These drugs reduce the risk of a clot forming on a plaque, but they increase the risk of bleeding, so the balance of benefit and risk must be assessed for each patient.

Surgery and procedures

When an artery is severely narrowed, surgery can restore blood flow. In balloon angioplasty, a catheter with a deflated balloon is threaded into the narrowed coronary artery. The balloon is inflated to push the plaque against the wall, and a stent, a small mesh tube, is often left in place to keep the artery open. In coronary artery bypass grafting, a healthy vessel taken from another part of the body is joined above and below the blockage, so blood flows around it.

KEY IDEA

Drugs that lower blood pressure reduce the load on the heart and the risk of artery damage. Drugs that lower cholesterol or prevent clots reduce the risk of blockage. Surgery restores flow once the blockage already exists.

Quick checks

QUICK CHECK

Which step is the earliest in the development of atherosclerosis?

- Formation of a blood clot on the plaque
- Narrowing of the lumen by fatty plaque
- Damage to the endothelium
- Rupture of an aneurysm

Atherosclerosis begins when the endothelial lining is damaged, which allows cholesterol and white blood cells into the wall. Clots and narrowing come later. Students often pick narrowing, because it is the most visible result, but it is not the first step.

QUICK CHECK

A person has a mass of 95 kg and a height of 1.70 m. What is their BMI?

- 55.8
- 25.9
- 32.9
- 0.56

BMI = mass ÷ height², so 95 ÷ (1.70 × 1.70) = 95 ÷ 2.89 = 32.87, which rounds to 32.9 and falls in the obese range. The wrong answers come from dividing by height rather than height squared, or by dividing in the wrong order.

QUICK CHECK

Why does smoking raise blood pressure?

- Carbon monoxide makes the blood thicker
- Nicotine makes the heart beat more slowly
- Nicotine releases adrenaline, which raises heart rate and constricts arteries
- Smoke causes the kidneys to hold water

Nicotine stimulates adrenaline secretion, increasing heart rate and narrowing arteries. Both raise blood pressure. Carbon monoxide reduces the oxygen the blood carries rather than making it thicker, which is a common confusion.

QUICK CHECK

Which drug type works by stopping the liver from making cholesterol?

- Beta blockers
- Diuretics
- Statins
- Aspirin

Statins inhibit HMG-CoA reductase, the enzyme that makes cholesterol in the liver. Beta blockers lower heart rate, diuretics remove water and salt, and aspirin reduces platelet stickiness. Those three work on different targets, which is why they are often confused.

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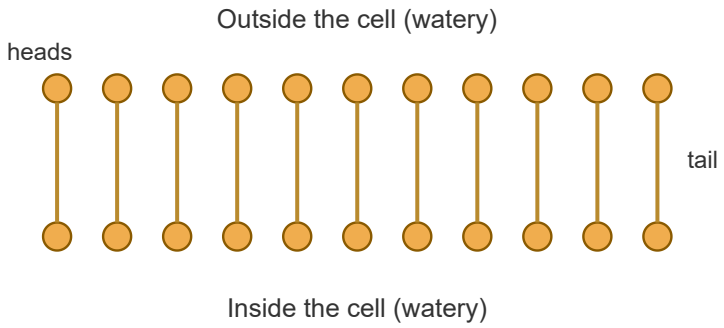
Membranes and transport

Every cell is enclosed by a membrane that decides what gets in and what gets out. This topic explains how the membrane is built from phospholipids and proteins, and how substances cross it by passive movement, by using energy, or by being packed into vesicles.

1. Why membranes form a bilayer

A phospholipid has two very different ends. The head is a phosphate group that is charged and attracted to water, so it is hydrophilic. The two tails are long fatty acid chains with no charge, so they are hydrophobic and avoid water. When phospholipids are placed in water they arrange themselves so that the heads face the watery solution on each side and the tails point inwards, away from it. The result is a double layer, or bilayer, with a water-hating core.

This arrangement is the reason membranes work as barriers. The hydrophobic core stops most charged ions and polar molecules from simply dissolving through the membrane, while small non-polar molecules can slip between the tails.



- **Phospholipid:** a molecule with a polar, hydrophilic phosphate head and two non-polar, hydrophobic fatty acid tails.
- **Phospholipid bilayer:** the double layer of phospholipids that forms the basic framework of every cell membrane.

2. The fluid mosaic model

Early models were simpler. The Gorter and Grendel model proposed a single phospholipid bilayer, but it could not explain how a membrane holds together its whole cell contents. The Davson-Danielli sandwich model placed a layer of protein on each face of a phospholipid core. This explained the membrane's strength but implied that all membranes look alike and that the proteins sit only on the surface.

The fluid mosaic model of Singer and Nicolson replaced these ideas. The membrane is a fluid sheet of phospholipids in which proteins are scattered like tiles in a mosaic. Phospholipids and many proteins can move sideways within their own layer, which is why the membrane is described as fluid. The model was supported by freeze-fracture images, which split the membrane down its middle and showed protein particles set into a smooth

matrix. It was also supported by experiments in which the membrane proteins of a mouse cell and a human cell mixed over time in a hybrid cell.

KEY IDEA

The Davson-Danielli model has proteins only on the surface and suggests a uniform structure. The fluid mosaic model has proteins that are embedded within the bilayer, and it allows the membrane to be fluid and varied. Both models have a phospholipid bilayer, so the difference lies in where the proteins are and how they move.

3. The parts of the membrane and what they do

Membrane proteins come in two broad groups. Integral proteins are embedded in the hydrophobic core of the bilayer, and many of them span the whole membrane. Peripheral proteins are not inside the bilayer at all. They are held loosely on its surface, often attached to the integral proteins. Among the integral proteins, channel proteins form water-filled pores that let polar molecules and ions through, while carrier proteins bind a specific molecule and change shape to move it across.

Cholesterol sits between the phospholipid tails. At warm temperatures it stops the phospholipids from moving so freely, which makes the membrane less fluid. At cool temperatures it stops the tails from packing tightly, which keeps the membrane fluid. Cholesterol is therefore a buffer that keeps the membrane working over a range of temperatures, and it also reduces the leakage of small polar molecules.

Carbohydrates are attached on the outer surface of the membrane. Where a carbohydrate is bound to a lipid this forms a glycolipid, and where it is bound to a protein it forms a glycoprotein. Together they make up the glycocalyx. These carbohydrate chains act as recognition markers, letting cells identify one another and receptor sites detect chemical signals.

Component	What it is	Main role
Phospholipid bilayer	Two layers of phospholipids	Forms a selectively permeable barrier
Integral protein	Embedded in the bilayer, often spanning it	Channels and carriers for transport
Peripheral protein	Loosely attached to the surface	Support and signalling
Cholesterol	Sits between the tails	Keeps fluidity steady at different temperatures
Glycolipid and glycoprotein	Carbohydrate bound to lipid or protein	Cell recognition and receptor sites

COMMON MISTAKE

Do not describe the proteins in the membrane as floating freely in a sea of lipid with no structure. The bilayer is organised: heads face water, tails face inwards, and the proteins are arranged around that core.

4. Passive transport: diffusion

Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration, down a concentration gradient. It happens because particles move randomly, so there are more movements away from a crowded area than into it. It needs no energy from the cell, which is why it is called passive.

Small non-polar molecules such as oxygen and carbon dioxide, and hydrophobic molecules such as benzene, pass straight through the phospholipid bilayer. Charged ions and large polar molecules cannot, because the hydrophobic core repels them. The rate of diffusion depends on the temperature (faster when hotter, because particles have more kinetic energy), the size of the concentration gradient, the surface area available, the size of the particles (smaller particles move faster), and the medium (diffusion is fastest in gases and slowest in solids).

Fick's law combines three of these factors for an exchange surface such as the lining of the alveoli: rate of diffusion $\propto$ (surface area × difference in concentration) ÷ thickness of the surface.

The surface available for exchange depends on an organism's size, as the surface area to volume ratio of a cube shows.

5. Passive transport: facilitated diffusion

Some substances are polar or charged, so they cannot cross the bilayer on their own, but they still move down their gradient. Facilitated diffusion uses a membrane protein to do this. Channel proteins open a water-filled route for ions or polar molecules. Carrier proteins bind the molecule on one side, change shape, and release it on the other side. Facilitated diffusion is still passive: it goes from high to low concentration and needs no ATP. It is faster than simple diffusion for the molecules it serves, but it is limited by the number of proteins available.

The cell can speed up or slow down the transport by changing how many carrier or channel proteins it places in the membrane. An example is glucose leaving liver cells into the blood through carrier proteins, which takes place without any energy input.

6. Osmosis and water potential

Osmosis is the net movement of water molecules across a partially permeable membrane. Water moves from a region where its concentration is higher to a region where it is lower. A dilute solution has a high concentration of free water, and a concentrated solution has a low one, because the dissolved solute takes up space and attracts some of the water. Water therefore moves from the dilute side to the concentrated side. Its movement can be described in terms of water potential, which is higher in dilute solutions and lower in concentrated ones.

What happens to a cell depends on the solution around it. In an isotonic solution the water movement in and out is balanced and the cell keeps its shape. In a hypotonic solution water enters: an animal cell may swell and burst, while a plant cell becomes turgid because the cell wall stops it from bursting. In a hypertonic solution water leaves: an animal cell shrinks, and a plant cell's cytoplasm pulls away from the wall, which is called plasmolysis.

Predicting what happens to a plant cell in a solution

1. Identify the water potential of the solution compared with the cell. A cell's contents usually have a lower water potential than pure water.
2. If the solution is more dilute than the cell (hypotonic), water moves in by osmosis, the vacuole fills, and the cell becomes turgid against the wall.
3. If the solution is more concentrated than the cell (hypertonic), water moves out, the vacuole shrinks, and the cytoplasm pulls away from the wall. This is plasmolysis.
4. Explain each step by pointing to the direction of the water potential gradient across the partially permeable membrane.

7. Active transport

Active transport moves molecules against their concentration gradient, from low concentration to high concentration. This is uphill, so the cell must supply energy. The energy usually comes from ATP. The substance to be moved binds to a carrier protein, ATP is used to change the shape of the protein, and the protein releases the substance on the other side of the membrane. Because the process is directed, it moves in one direction only. Active transport is needed when the cell must keep a high concentration of a substance inside, such as potassium ions in cells, or must remove a substance that is at a lower concentration outside, such as absorbing mineral ions from the soil into root hair cells.

KEY IDEA

Active transport always needs energy and always uses carrier proteins. Facilitated diffusion uses protein but no energy, and always runs down the gradient.

Feature	Facilitated diffusion	Active transport
Direction	Down the concentration gradient	Against the concentration gradient
Energy needed	No	Yes (ATP or an ion gradient)
Membrane proteins	Channel or carrier proteins	Carrier proteins
Example	Glucose leaving liver cells	Mineral ions entering root hair cells

8. Bulk transport: endocytosis and exocytosis

Very large molecules, or many molecules at once, are moved across the membrane in vesicles. In endocytosis the membrane folds inwards around material outside the cell, and a new vesicle pinches off inside the cytoplasm. In exocytosis a vesicle, often one made by the Golgi apparatus, moves to the membrane and fuses with it, releasing its contents outside. Both processes use energy from ATP and depend on the membrane's fluidity, which lets it bend and fuse.

EXAM TIP

When you name a process in an exam, match the direction to the name: endocytosis brings material in, exocytosis sends it out.

9. Summary

Process	Direction	Energy	Protein needed	Example
Diffusion	High to low concentration	No	No	Oxygen moving into cells
Facilitated diffusion	High to low concentration	No	Yes	Glucose entering cells
Osmosis	Water from dilute to concentrated	No	Water passes through the bilayer or channels	Water entering kidney tubules
Active transport	Low to high concentration	Yes (ATP)	Yes	Mineral ions into root hair cells
Endocytosis and exocytosis	Into or out of the cell	Yes (ATP)	Vesicle formation	Secretion from secretory cells

10. Quick checks

QUICK CHECK

Which molecule is most likely to cross a phospholipid bilayer by simple diffusion?

- Sodium ions
- Glucose
- Oxygen
- Amino acids

Oxygen is small and non-polar, so it can dissolve in the hydrophobic core of the bilayer. Sodium ions are charged and glucose is a large polar molecule, so both need a protein. Amino acids are polar and cannot pass the core unaided.

QUICK CHECK

A red blood cell is placed in a very dilute salt solution. What is the most likely result?

- It shrinks as water leaves
- It swells and may burst as water enters
- It stays the same size because the solution is isotonic
- It becomes turgid because it has a cell wall

A dilute solution is hypotonic to the cell, so water enters by osmosis. Animal cells have no wall to resist the increase in volume, so they swell and can burst. The turgid answer applies to plant cells, which have a wall.

QUICK CHECK

Which statement best describes active transport?

- Movement down a gradient using a channel protein
- Movement of water through a partially permeable membrane
- Movement against a gradient, using carrier proteins and energy from ATP
- Bulk movement of solid particles through a single pore

Active transport moves substances against the gradient, so it needs ATP and carrier proteins that change shape. Channel-based movement down the gradient is facilitated diffusion, and osmosis concerns water only.

QUICK CHECK

Why does cholesterol help keep a membrane working across a range of temperatures?

- It makes the bilayer fully solid at low temperatures
- It pumps ions across the membrane
- It restrains phospholipid movement when warm and prevents tight packing when cool
- It attaches carbohydrates to the outer surface

Cholesterol stops the membrane becoming too fluid at high temperatures and too rigid at low temperatures. It does not make the membrane solid, pump anything, or attach carbohydrates; those roles belong to other molecules.

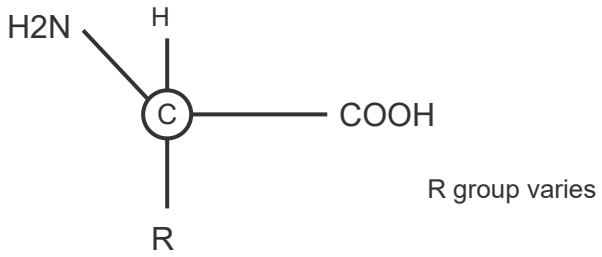
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Proteins and DNA

Proteins do most of the work in a cell, and DNA holds the instructions for building them. This topic follows the story from amino acids, through the folding of a polypeptide into a working protein, to the double helix of DNA and the way it copies itself before each cell division.

1. Amino acids

Every amino acid has the same core. A central carbon atom carries an amine group (-NH2), a carboxyl group (-COOH), a hydrogen atom, and a variable side chain called the R group. The R group is what distinguishes one amino acid from another. Its size, charge and polarity decide how the amino acid behaves, and so how the finished protein folds.



2. Forming a polypeptide

Amino acids are joined by a peptide bond. The carboxyl group of one amino acid reacts with the amine group of the next, and a molecule of water is removed in a condensation reaction. The bond formed is the C-N link shown in the diagram. A chain of amino acids is a polypeptide, and a chain with two units is a dipeptide. Adding water back in breaks the peptide bond again, in a hydrolysis reaction.

The order of amino acids in the chain is the primary structure. That order is set by the sequence of bases in the DNA gene that codes for the protein, so DNA controls which protein is made.

3. How a protein folds

A chain of amino acids does not stay straight. It folds into a shape that is held in place by several kinds of bond, and each level of folding builds on the one below it.

- **Primary structure:** the sequence of amino acids joined by peptide bonds.
- **Secondary structure:** local folding of the chain into an alpha helix or a beta pleated sheet. Both are held together by hydrogen bonds between the C=O group of one amino acid and the N-H group of another, a few residues further along the chain.

- **Tertiary structure:** the overall three-dimensional shape of a single polypeptide, maintained by bonds between R groups. These are hydrogen bonds, ionic bonds between charged R groups, disulfide bridges between cysteine residues, and hydrophobic interactions.
- **Quaternary structure:** two or more polypeptide chains held together to form one functional protein.

The shape of the final protein is what matters. Hydrophilic R groups tend to sit on the outside of a soluble protein where they can meet water, and hydrophobic R groups tend to be buried in the middle. A protein's function depends on its shape, especially on the shape of any active site or binding site.

KEY IDEA

Primary structure is fixed by the gene. Secondary and tertiary structure are determined by the bonds that the amino acid sequence allows. Changing just one amino acid can change which R groups are close together, and that can change the whole shape.

4. Fibrous and globular proteins

Proteins fall into two broad shape types. Fibrous proteins are long and repetitive, with polypeptides lying parallel to one another. They are usually insoluble and strong, and they are good for structure. Collagen, found in tendons, ligaments, bone and skin, is a fibrous protein with three polypeptides wound into a triple helix. Repeated small amino acids such as glycine let the chains pack tightly, and many hydrogen bonds hold the helix together. This gives collagen high tensile strength.

Globular proteins are folded into compact, roughly spherical shapes. Their hydrophilic R groups face outwards, which makes them soluble in water. Their precise shape allows them to bind other molecules, so globular proteins act as enzymes, carriers and transport proteins. Haemoglobin is a globular protein made of four polypeptide chains, two alpha and two beta, each holding one haem group that contains an iron ion. Each haem group can bind one oxygen molecule, so one haemoglobin molecule carries four.

Feature	Fibrous protein (e.g. collagen)	Globular protein (e.g. haemoglobin)
Overall shape	Long and rod-like	Compact and roughly spherical
Solubility	Insoluble	Soluble
Amino acid pattern	Repetitive, often small R groups	Varied, with hydrophilic R groups outside
Main job	Structural support	Transport, binding and catalysis

5. Denaturation

Heat, extreme pH, high salt concentration and mechanical agitation all disturb the bonds that hold a protein in shape. The protein unfolds, its active or binding site is lost, and it stops working. Denaturation affects secondary, tertiary and quaternary structure, but the primary structure, the sequence of peptide bonds, is not broken. This is why a denatured protein is not destroyed in the chemical sense, but it no longer functions.

COMMON MISTAKE

Denaturation does not break peptide bonds. Breaking a peptide bond is hydrolysis, which needs water and an enzyme or strong acid, and it is a different process. Denaturation breaks the weaker bonds that shape the protein.

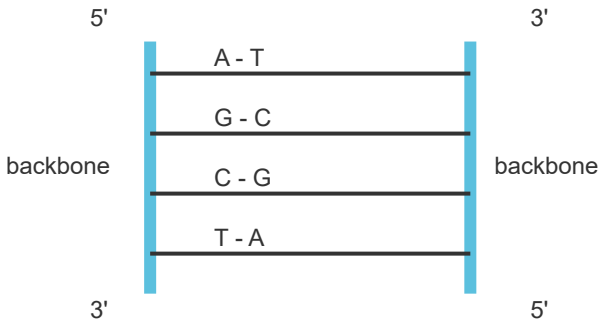
Enzymes are globular proteins, so temperature and pH change their rate of reaction by changing the shape of the active site.

6. The structure of DNA

DNA is a polymer of nucleotides. Each nucleotide has three parts: a phosphate group, a pentose sugar, and a nitrogen-containing base. In DNA the sugar is deoxyribose and the bases are adenine (A), thymine (T), cytosine (C) and guanine (G). Adenine and guanine are purines, which have a double-ring structure. Cytosine and thymine are pyrimidines, which have a single ring. RNA differs in having ribose instead of deoxyribose and uracil (U) in place of thymine.

Nucleotides link together by condensation reactions. The phosphate attached to one sugar joins to the third carbon of the sugar in the next nucleotide, forming a phosphodiester bond. This makes a sugar-phosphate backbone with the bases projecting from it. Each chain has a 5' end, which carries a phosphate, and a 3' end, which carries a free hydroxyl group.

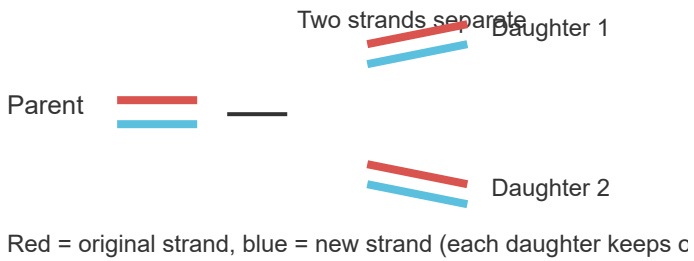
In DNA two chains run side by side in opposite directions, so they are antiparallel. The bases of one chain pair with those of the other by hydrogen bonds. Complementary pairing is strict: A pairs only with T, using two hydrogen bonds, and C pairs only with G, using three. Because every purine pairs with a pyrimidine, the two strands have the same width all the way along, and the whole molecule twists into a regular double helix with about ten base pairs per turn.



7. Semi-conservative DNA replication

Before a cell divides, its DNA must be copied so that each daughter cell receives a full set. Replication starts when the enzyme helicase breaks the hydrogen bonds between the base pairs, so the double helix unwinds and the two strands separate. Each strand now acts as a template. Free nucleotides in the nucleus pair with exposed bases by complementary base pairing: A with T, and C with G.

DNA polymerase then joins the neighbouring nucleotides along each new chain, forming phosphodiester bonds through condensation. The result is two daughter molecules. Each one contains one original strand and one newly made strand. This is why the process is called semi-conservative: half of each molecule is conserved from the parent.



Red = original strand, blue = new strand (each daughter keeps c

8. The Meselson and Stahl experiment

Three replication models were possible. In the conservative model the original double helix stays intact and a completely new copy is made. In the semi-conservative model each daughter keeps one old strand. In the dispersive model the old DNA is broken into pieces and mixed with new material, so every strand is part old and part new.

Meselson and Stahl tested these models by growing bacteria in medium containing the heavy isotope nitrogen-15, so their DNA became dense. They then moved the bacteria into medium with the light isotope nitrogen-14. DNA samples were spun in a density gradient, which separates molecules by density. Heavy DNA sits lowest, light DNA sits highest, and hybrid DNA, with one heavy and one light strand, sits between them.

Model	After one generation	After two generations
Conservative	Half heavy, half light (no intermediate band)	Heavy and light bands only
Semi-conservative	One intermediate band only	Intermediate and light bands in equal amounts
Dispersive	One intermediate band only	One intermediate band only

Compare the predictions of the three models generation by generation.

Showing that replication is semi-conservative

1. Grow the bacteria in heavy nitrogen for several generations so that all their DNA is heavy.
2. Transfer them to light nitrogen. After one generation there is only an intermediate band. The conservative model is ruled out, because it would give a heavy and a light band.
3. After a second generation there is an intermediate band and a light band. The dispersive model is ruled out, because it would give only intermediate DNA.

4. Conclude that each new molecule contains one original strand and one newly made strand, which is the semi-conservative model.

EXAM TIP

In an exam, state the band pattern for each model after each generation. Then say which model each result rules out. Simply naming the semi-conservative model is not enough.

9. Quick checks

QUICK CHECK

What type of bond joins two amino acids together in a polypeptide?

- Hydrogen bond
- Disulfide bridge
- Peptide bond
- Phosphodiester bond

Amino acids are linked by a peptide bond, formed by condensation between the carboxyl group of one and the amine group of the next. Hydrogen bonds and disulfide bridges hold the folded protein together rather than linking the chain, and phosphodiester bonds join nucleotides in DNA.

QUICK CHECK

Which pair of bases is joined by three hydrogen bonds in DNA?

- Adenine and thymine
- Adenine and guanine
- Thymine and cytosine
- Cytosine and guanine

Cytosine and guanine form three hydrogen bonds, while adenine and thymine form two. Adenine and guanine are both purines, so they cannot pair with each other, and thymine and cytosine are both pyrimidines.

QUICK CHECK

In the Meselson and Stahl experiment, what result after one generation in light nitrogen supports semi-conservative replication?

- Only a light band
- Equal heavy and light bands
- A single intermediate band
- Only a heavy band

Semi-conservative replication gives each new molecule one heavy and one light strand, which forms an intermediate band. A conservative model would have given half heavy and half light, and a light-only or heavy-only result would mean no replication had occurred.

QUICK CHECK

Which change would most directly denature a protein?

- Breaking the peptide bonds with water
- Adding a new amino acid to the end of the chain
- Heating it to a high temperature
- Increasing the length of the DNA gene

Heat disrupts the hydrogen, ionic and disulfide bonds that hold the folded shape, so the protein unfolds and loses function. Breaking peptide bonds is hydrolysis, not denaturation, and the other two options do not change the existing protein's shape.

AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

Gene expression and genetics

Every cell in your body carries the same DNA, yet a nerve cell and a liver cell look and behave very differently. This page explains how cells switch genes on and off, how one gene can produce more than one protein, how chemical changes to DNA and its packaging alter gene activity without changing the base sequence, and why the characteristics we observe depend on both genes and the environment.

1. Controlling gene expression

Gene expression means turning the information in a gene into a working product. For a protein-coding gene this happens in two stages. In transcription, a section of DNA is copied into a molecule of messenger RNA (mRNA) inside the nucleus. In translation, the mRNA travels to a ribosome in the cytoplasm and is read in codons to build a polypeptide chain. Both stages can be regulated, but the commonest point of control is transcription, because switching a gene off at this stage stops the whole chain of events before any protein is made.

Different cells express different combinations of genes. A cell that makes large amounts of digestive enzymes has many copies of the enzyme mRNA, while a muscle cell is rich in the proteins that contract. Because the proteins present in a cell, and how much of each, largely set its structure and function, the cell's type is effectively decided by which genes are active. Every cell has the full set of genes, so specialisation is about selection, not loss.

- **Transcription:** copying one strand of a gene's DNA into pre-mRNA, catalysed by RNA polymerase.
- **Translation:** reading mRNA codons at a ribosome to join amino acids in the order specified.
- **Transcription factor:** a protein that binds to DNA near a gene and either helps or blocks RNA polymerase from starting transcription.
- **Cell specialisation:** the permanent change in cell structure and function that follows from switching on a particular set of genes.

Transcription factors

The most common way a cell controls a gene is by switching its transcription on or off. Transcription factors do this. Some bind to the promoter region, the stretch of DNA just before a gene where RNA polymerase normally attaches, and either recruit the polymerase or prevent it from binding. Others bind to enhancer sequences, which may lie some distance from the gene. These enhancer-binding factors change the local structure of chromatin, the DNA-protein complex in the nucleus, so that the gene is either more or less accessible to the polymerase.

Open chromatin, where the DNA is loosely packed, is linked to active genes. Closed or condensed chromatin is linked to inactive genes. Controlling transcription is an efficient strategy: one mRNA molecule can be translated many times by ribosomes, so a single switch at the mRNA stage produces a large number of protein molecules.

KEY IDEA

Transcription is the most efficient control point because one mRNA molecule can be read by many ribosomes at once. Switching transcription off therefore cuts protein production far more than blocking a single translation event would.

Cell specialisation in practice

Chemical signals such as hormones trigger specialisation. A hormone binds to a receptor, the hormone-receptor complex moves into the nucleus, and it activates particular genes. Those genes are transcribed into mRNA, the mRNA is translated, and the new protein changes what the cell does. Because the change is carried by proteins that persist, and because the cell passes its gene-expression pattern to daughter cells at division, the specialised state is maintained over time.

EXAM TIP

In exam answers, describe specialisation as a chain: signal, receptor, activated gene, transcription, translation, protein, changed cell. Examiners look for the order of steps, not just the words "gene switched on".

2. One gene, many proteins

If each gene made exactly one protein, the number of proteins in a cell would equal the number of genes. In reality a single gene can give rise to several different proteins. The explanation lies in what happens to the RNA transcript after transcription, not in the DNA itself.

How one gene gives different mRNAs

1. Transcription of the gene produces a primary RNA transcript (pre-mRNA) that contains both exons and introns. Exons are the sequences that will end up in the final message; introns are the sequences that will be removed.

2. Splicing removes the introns from the transcript. Spliceosomes, which are complexes of proteins and small RNAs, carry out this cutting and rejoining.

   Only the exons carry codons for amino acids, so the introns must be removed before translation can produce a correct polypeptide.

3. The exons are not always joined in the same combination. In some cells certain exons are left out, and in others different exons are kept. This is alternative splicing.

4. Each combination gives a different mRNA, with a different sequence and length. A longer mRNA contains more codons, so the polypeptide it encodes has more amino acids.

5. Different amino acid sequences form different bonds during folding, so the same gene can produce proteins that fold into different three-dimensional shapes and perform different functions.

Shape, not sequence alone, determines what a protein does, so changing which exons are joined changes function.

COMMON MISTAKE

Alternative splicing does not change the DNA. Students often write that the gene "mutates" or "changes" to make new proteins. The gene sequence stays the same; only the processing of its transcript differs.

3. Epigenetics: control beyond the base sequence

Epigenetics is the study of control over gene activity that does not come from the order of bases in the DNA. The same base sequence can be active in one cell and silent in another. Two main systems do this: DNA methylation and histone modification. Both respond to the environment, and both can be inherited when cells divide, which is why they matter for development and for the phenotype of an organism.

RNA splicing also counts as an epigenetic process in this sense, because it changes which mRNAs and proteins are produced from the same genetic code.

DNA methylation

A methyl group (-CH3) can be added to a cytosine base when that cytosine sits next to a guanine in the DNA chain, with a phosphate linking them. The enzyme that adds it is a DNA methyltransferase, which uses a methyl donor molecule. Once a methyl group is attached, the local DNA is less available to the transcription machinery, so the gene tends to be silenced. Removing the methyl group, which is demethylation, allows the gene to be transcribed again.

Methylation also changes the histones that the DNA wraps around, so one chemical change can affect gene activity in more than one way.

Histone modification

DNA is wound around histone proteins, which are rich in positively charged amino acids such as lysine. The DNA is negatively charged because of its phosphate groups, so the two are attracted to each other. The DNA-histone complex is chromatin, and chromatin packs into chromosomes. How tightly the DNA is wound determines whether genes can be read. Densely packed heterochromatin hides genes from the transcription machinery. Loosely packed euchromatin exposes them.

Chemical groups added to histones change how tightly they hold the DNA:

- **Acetylation:** an acetyl group is added to a lysine on the histone. The acetyl group neutralises the positive charge, so the histone holds DNA less tightly. The chromatin opens and genes in that region become available for transcription.
- **Methylation (of histones):** a methyl group is added to a lysine. This is usually linked to gene silencing, and sometimes to whole-chromosome silencing, because the nucleosomes pack more tightly and transcription factors cannot reach the DNA.

KEY IDEA

Acetylation tends to switch genes on by loosening chromatin. Methylation of DNA and of histones tends to switch genes off by tightening it. Both act on the same question: can the transcription machinery reach this gene?

Epigenetic control in human development: haemoglobin

Haemoglobin shows epigenetic control clearly. Adult haemoglobin contains two alpha and two beta globin chains. Fetal haemoglobin contains two alpha and two gamma globin chains. During development, the gamma genes are switched on first and then switched off just before and after birth, while the beta genes are switched on. The switch from gamma to beta production is not caused by any change in the genes themselves.

Histone acetylation appears to activate the fetal gamma genes. DNA methylation seems to silence them around birth, and histone methylation complements this by reinforcing the silencing. The graph shows how the proportion of each globin type changes with time from conception through to the months after birth.

EXAM TIP

When asked about epigenetic control in development, name the mechanism (acetylation or methylation), say what it does to chromatin, and link it to which gene is switched on or off at a given stage.

4. Phenotype, genotype and environment

A phenotype is the observable characteristic of an organism. It is not fixed by genotype alone. The genotype sets a potential, and the environment determines how much of that potential is reached. Height is the standard example: a person may carry genes for tall stature but still fail to reach full height if they are malnourished in childhood, because bone and muscle growth depend on protein and calcium. Health, which can also change gene expression, affects height in the same way.

Environmental effects can begin before birth. Poor maternal nutrition can leave a fetus short of nutrients needed for growth, and smoking reduces the oxygen supply, which can affect the growth of the body and brain. Twin studies help to separate genetic and environmental effects. Identical twins share essentially all their DNA, so differences between them are attributed to environment. Non-identical twins share about half their DNA on average, like ordinary siblings, but tend to share a more similar environment, which is why they are a useful comparison.

Skin and hair colour

Melanin is the dark pigment in skin and hair. It is produced by melanocytes, which are activated by melanocyte-stimulating hormone (MSH). MSH binds to receptors on melanocytes. Melanocytes package melanin into melanosomes, which are passed to surrounding skin and hair cells and gather around the nucleus, shielding the DNA from ultraviolet (UV) light. Individuals with more receptors and more melanin therefore have darker skin and greater protection against sunburn.

UV light increases the amount of MSH and the number of MSH receptors, making melanocytes more active. This is an environmental effect on a genotype-defined pathway. Hair can appear to lighten with UV exposure because UV damages melanin and other proteins in hair cells, even though more pigment is being made.

5. Continuous and discontinuous variation

Some characteristics are controlled almost entirely by genotype. Blood group is an example: the environment has little or no effect, and the phenotypes fall into separate categories with no overlap. Such characteristics are controlled by genes at a single locus and show discontinuous variation. A bar chart of their frequencies has gaps between the bars.

A characteristic controlled by one gene can be followed from parents to offspring with a genetic diagram such as a Punnett square.

Other characteristics, such as human height, show continuous variation. Their phenotypes form a smooth range with no clear categories. These characteristics are usually controlled by many genes at different loci, a pattern called polygenic inheritance, and they are also affected by the environment, either directly or by changing gene expression. When the frequency of each height category is plotted, the result is a bell-shaped curve, which indicates normally distributed data.

> **KEY IDEA**
>
> Discontinuous variation: single locus, genotype dominant, discrete categories, no overlap. Continuous variation: many loci (polygenic), environment also matters, a range of values, normal distribution.

Polygenic inheritance

In polygenic inheritance, alleles at several loci interact to determine one characteristic. Each allele has only a small effect, and the combined effects of many alleles produce the individual's phenotype. Eye colour is a good example. Several loci influence how much pigment is in the iris. Brown eyes contain a lot of pigment, which absorbs light and makes them look dark. Blue eyes contain little pigment, so light is reflected from the iris. Intermediate pigment levels give the range of colours in between.

Adding more loci, and then the effect of the environment, turns a few separate classes into a bell-shaped curve.

QUICK CHECK

A characteristic shows continuous variation and a bell-shaped distribution. Which explanation is most likely?

- It is controlled by a single gene with two alleles and no environmental effect.
- It is controlled by many genes at different loci and is also affected by the environment.
- It is caused by a single mutation that changes the amino acid sequence of one protein.
- It is controlled by a single locus, and the phenotypes fall into separate groups.

Continuous variation with a normal distribution is produced by polygenic inheritance, where several loci each contribute a small effect, and the environment adds further variation. A single gene with no environmental effect gives discontinuous variation instead, which is the most common wrong answer.

QUICK CHECK

Which change would most directly switch a gene off by making its DNA less accessible to transcription?

- Acetylation of histones at the gene's region
- Demethylation of the cytosine bases in the gene
- Methylation of cytosine bases in the gene's DNA
- Splicing out an intron from the pre-mRNA

Methylation of cytosine next to guanine is linked to gene silencing, because the methyl group reduces transcription. Acetylation and demethylation both tend to activate genes. Splicing acts after transcription and does not change DNA accessibility, so it is the commonest wrong answer.

QUICK CHECK

A single gene produces three different proteins in three different cell types. What is the most likely reason?

- The gene sequence has mutated differently in each cell type.
- Each cell type has a different number of copies of the gene.
- Different combinations of exons are joined during splicing, giving different mRNAs.
- Each protein is produced by a different chromosome that carries the gene.

Alternative splicing joins the exons in different combinations, giving mRNAs of different length and sequence, and therefore different proteins. The DNA is the same in every cell, so mutation is not the explanation. Gene copy number is normally the same in each diploid cell.